SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I **Gadi Reddy Prakash / 192365026** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

G. Reddy Prakash
Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1. **Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.**

    a. **Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.**
    b. **Explain how an authorized member, aware of the encoding rule, would retrieve the original message.**

Sol:-

Scenario

A student organization wants to securely share confidential event details with its core members. To prevent unauthorized people from reading the message, they use a Caesar Cipher, which is one of the oldest classical encryption techniques. During testing, the message "MEET AT NOON" is encrypted using a fixed shift of 4 positions.

Introduction

The **Caesar Cipher** is a substitution cipher invented by **Julius Caesar**. In this technique, every letter in the plaintext is replaced by another letter that is a fixed number of positions ahead in the English alphabet.
For this scenario,

- **Plaintext:** MEET AT NOON
- **Shift Value:** +4
- **Encryption Method:** Caesar Cipher

The encryption process protects the original message during communication.

Diagram 1: Caesar Cipher Working

Caesar Cipher

Plaintext
"MEET AT NOON"
↓
Shift every letter
by +4 positions
↓
Ciphertext
"QIIX EX RSSR"

(a) Encryption Process (Shift = +4)

Each alphabet is shifted **4 positions forward**.

Plain Letter	Shift (+4)	Cipher Letter
M	Q	Q
E	I	I
E	I	I
T	X	X
A	E	E
T	X	X
N	R	R
O	S	S
O	S	S
N	R	R

Step-by-Step Encryption

Position	Plain Letter	Shift	Cipher Letter
1	M	+4	Q
2	E	+4	I
3	E	+4	I

Position	Plain Letter	Shift	Cipher Letter
4	T	+4	X
5	Space	-	Space
6	A	+4	E
7	T	+4	X
8	Space	-	Space
9	N	+4	R
10	O	+4	S
11	O	+4	S
12	N	+4	R

Encrypted Message
Plaintext
MEET AT NOON
↓
Ciphertext
QIIX EX RSSR

Alphabet Mapping Table

Plain	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**
Cipher	E	F	G	H	I	J	K	L	M	N	O	P	Q
Plain	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
Cipher	R	S	T	U	V	W	X	Y	Z	A	B	C	D

Diagram 2: Alphabet Shift
Plain Alphabet

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Shift +4

E F G H I J K L M N O P Q R S T U V W X Y Z A B C D

(b) Decryption Process
An authorized member knows that the sender used a **shift value of +4**.
To recover the original message, every ciphertext letter is shifted **4 positions backward**.

Step-by-Step Decryption

Cipher Letter	Shift (-4)	Plain Letter
Q	M	M
I	E	E
I	E	E
X	T	T
E	A	A
X	T	T
R	N	N
S	O	O
S	O	O
R	N	N

Recovered Message
QIIX EX RSSR

↓

MEET AT NOON
The receiver successfully retrieves the original message because both sender and receiver know the **same**

shift value (4).

Diagram 3: Encryption and Decryption Process

```
        Sender

   MEET AT NOON
       |
       ▼
   Caesar Cipher
   Shift +4
       |
       ▼
   QIIX EX RSSR
       |
──────────────Network──────────────
       |
       ▼
    Receiver
       |
   Shift −4
       ▼
   MEET AT NOON
```

Advantages of Caesar Cipher

Advantages	Description
Simple	Easy to understand and implement.
Fast	Encryption and decryption require very little computation.
Educational	Useful for learning basic cryptography concepts.

Limitations

Limitations	Explanation
Small Key Space	Only 25 possible shifts make brute-force attacks easy.
Low Security	Attackers can easily guess the shift value.
Frequency Analysis	Letter frequency reveals the plaintext quickly.

Applications

- Educational demonstrations of cryptography.
- Learning classical encryption techniques.
- Simple puzzles and games.
- Historical military communication.

Conclusion

The **Caesar Cipher** is one of the simplest classical encryption techniques. By shifting each alphabet **4 positions forward**, the plaintext **"MEET AT NOON"** becomes **"QIIX EX RSSR"**. An authorized receiver can decrypt the ciphertext by shifting each letter **4 positions backward**, successfully recovering the original message. Although the Caesar Cipher is useful for understanding the fundamentals of cryptography, it is not secure enough for modern communication because it is vulnerable to brute-force and frequency-analysis attacks.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEEFEAOCAIVDEN”.

a. **Determine the most probable original message by reversing the encryption process.**
b. **Critically evaluate why such a method is vulnerable in modern communication systems.**
c. **Propose one practical enhancement to improve the confidentiality of messages in this scenario.**

Scenario
A company still uses a **transposition-based encryption method** in its legacy communication system. The received ciphertext is:
WECRLTEERDSOEEFEAOCAIVDEN
The encryption technique uses **three levels (3 rails)**. The task is to recover the original message, evaluate its security weaknesses, and recommend an improvement.

Introduction
A **Rail Fence Cipher** is a classical **transposition cipher** that rearranges the positions of letters without changing the letters themselves. Unlike substitution ciphers, the original letters remain unchanged, but their order is altered according to a zigzag (rail fence) pattern.
In this scenario:

- **Cipher Technique:** Rail Fence Cipher
- **Number of Rails:** 3
- **Ciphertext:** WECRLTEERDSOEEFEAOCAIVDEN

Diagram 1: Rail Fence Encryption Process
Rail Fence Cipher

Plaintext Message
↓
Arrange letters in Zigzag
(3 Rails Pattern)
↓
Read row-by-row to obtain
Encrypted Message

(a) Determine the Original Message
The ciphertext is
WECRLTEERDSOEEFEAOCAIVDEN
To recover the original message, the receiver reconstructs the zigzag pattern using **3 rails** and fills the ciphertext row by row.

Step 1: Zigzag Pattern
W . . . E . . . C . . . R . . . L . . . T . . . E
. E . R . D . S . O . E . E . F . E . A . O . C .
. . A . . . I . . . V . . . D . . . E . . . N . .

Step 2: Read Zigzag Vertically
Reading the zigzag path gives
WEAREDISCOVEREDFLEEATONCE

Recovered Plaintext

Ciphertext	Original Message
WECRLTEERDSOEEFEAOCAIVDEN	**WE ARE DISCOVERED FLEE AT ONCE**

Diagram 2: Rail Fence Decryption

Ciphertext

WECRLTEERDSOEEFEAOCAIVDEN

↓

Fill the three rails

↓

Read in zigzag order

↓

WE ARE DISCOVERED FLEE AT ONCE

Step-by-Step Decryption Process

Step	Description
1	Identify that the cipher uses **3 rails**.
2	Draw the zigzag pattern.
3	Place ciphertext letters row by row into each rail.
4	Read letters following the zigzag path.
5	Recover the original plaintext.

Rail Fence Arrangement Table

Rail 1	W E C R L T E
Rail 2	E R D S O E E
Rail 3	A I V D E N

Following the zigzag order reconstructs the original message.

(b) Why is this Method Vulnerable in Modern Communication Systems?

Although the Rail Fence Cipher was useful historically, it is **not suitable for modern secure communication**.

Reasons

Vulnerability	Explanation
Small Key Space	The number of rails is usually very small, making brute-force attacks easy.
No Letter Substitution	Letters remain unchanged, only their positions change.
Pattern Recognition	Attackers can identify zigzag patterns using frequency analysis and trial methods.
No Authentication	Cannot verify whether the message has been modified.
No Confidentiality	Modern computers can decrypt the cipher within seconds.

Comparison with Modern Encryption

Feature	Rail Fence Cipher	AES Encryption
Security	Low	Very High
Key Size	Very Small	128/192/256 bits
Brute Force Resistance	Poor	Excellent
Used Today	No	Yes
Suitable for Sensitive Data	No	Yes

Diagram 3: Security Comparison

Legacy System

Message

↓

Rail Fence Cipher

│

Weak Security

│

Easy to Break

Modern System

Message
│
▼
AES Encryption
│
Strong Key
│
Highly Secure

(c) Practical Enhancement
A practical improvement is to replace the Rail Fence Cipher with the **Advanced Encryption Standard (AES-256)**.
Why AES?

- Uses a **256-bit encryption key**.
- Resistant to brute-force attacks.
- Provides strong confidentiality.
- Widely used in banking, cloud computing, military systems, and secure messaging.
- Approved as an international encryption standard.

Improved Communication Process
Sender
│
▼
Original Message
│
▼
AES Encryption
│
Encrypted Data
│
────────── Internet ──────────
│
Encrypted Data
│
▼
AES Decryption
│
Original Message

Benefits of AES

Benefit	**Description**
High Security	Protects sensitive information effectively.
Fast Performance	Suitable for both hardware and software implementation.
Global Standard	Trusted worldwide for secure communication.
Strong Key Management	Uses large keys that are computationally infeasible to crack.

Advantages of Modern Encryption over Rail Fence

Rail Fence Cipher	**AES Encryption**
Simple transposition	Complex mathematical encryption
Easy to crack	Extremely difficult to crack
No integrity protection	Can be combined with authentication mechanisms
Not suitable today	Used in modern communication systems

Conclusion
The given ciphertext **"WECRLTEERDSOEEFEAOCAIVDEN"** is successfully decrypted using the **3-Rail Rail Fence Cipher**, revealing the original message: